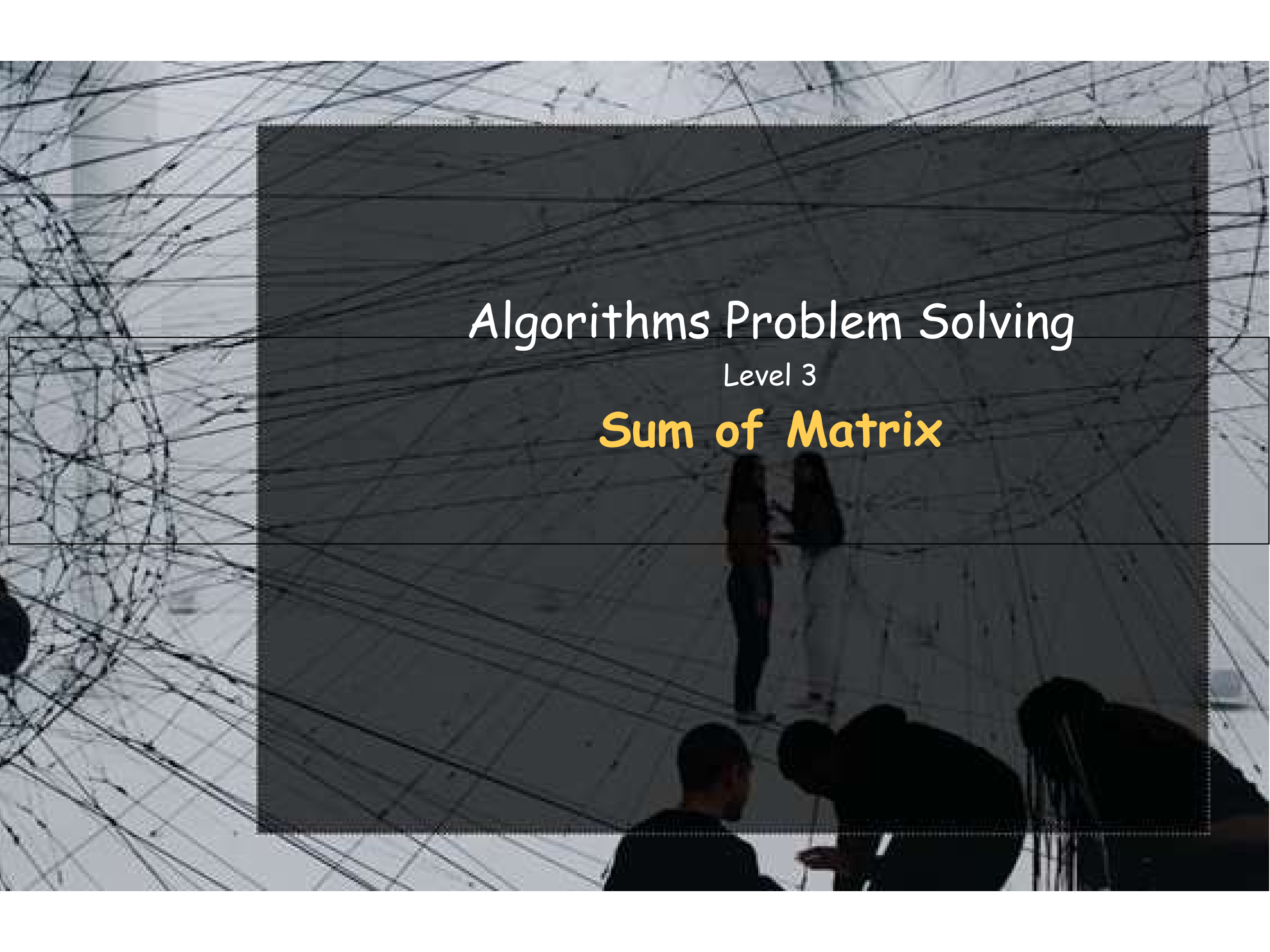


Algorithms Problem Solving

Level 3

Sum of Matrix



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Sum of Matrix

Definition

Target Letter Frequency Counter

-  Program counts occurrences of a selected character in a string.
-  It can perform both case-sensitive and case-insensitive counting.

Explanation

Case-Sensitive Matching

-  The program compares each character directly with the target.

-  For example, ``'m'`` matches ``'m'`` but not ``'M'``.

Case-Insensitive Matching

-  Both the current character and target character are normalized to the same case.

-  This allows ``'m'`` and ``'M'`` to be treated as the same letter.

Independent Counters

-  One counter records exact-case matches.

-  Another records matches regardless of case.

Procedures

- 1 Read the complete string.
- 2 Read the target character.
- 3 Perform a case-sensitive scan.
- 4 Perform a case-insensitive scan.
- 5 Display both counts.

💡 Example

String: Sara Alterkawi

Letter: 'a'

➡ Result

Letter 'a' Count = 3

Letter 'm' Or 'M' Count = 4

Note

- ◆ Case-insensitive comparison can be implemented by converting both values using ``tolower()``.
- ◆ Passing the string as ``const string&`` avoids an unnecessary copy.
- ◆ The target character itself should also be normalized in case-insensitive mode.

⊘ Warning

- ◆ Mixing ``cin >>`` and ``getline()`` may require handling the remaining newline in the input buffer.
- ◆ Direct comparison is case-sensitive.
- ◆ Character conversion should be performed consistently on both the target and the string characters.



Comparison

▨ Case-Insensitive Count

- ✓ Treats uppercase and lowercase forms as equivalent.
- ✓ Normalizes both characters before comparison.
- ✓ Counts the total frequency independent of case.

▤ Case-Sensitive Count

- ✓ Requires exact character equality.
- ✓ Distinguishes ``'a'`` from ``'A'``.
- ✓ Is appropriate when letter case carries meaning.

✦✦ Summary

- ✓ Searches a string for a selected character.
- ✓ Supports exact case-sensitive matching.
- ✓ Supports case-insensitive matching.
- ✓ Demonstrates how character normalization changes search behavior.

Thank you 😊